ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

Study Material

Binomial Theorem

(English Medium)

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BINOMIAL THEOREM

1. BINOMIAL EXPRESSION :

Any algebraic expression which contains two dissimilar terms is called binomial expression.

For example : $x - y,\ xy + \frac{1}{x},\ \frac{1}{z} - 1,\ \frac{1}{(x-y)^{1/3}} + 3$ etc.

2. Pascal's triangle :

A triangular arrangement of numbers as shown is called Pascal's triangle. The numbers give the binomial coefficients (or combinatorial coefficients) for the expansion of $(x + y)^n$. The first row is for n = 0, the second for n = 1, etc. Each row has 1 as its first and last number. Other numbers are generated by adding the two numbers immediately to the left and right in the row above.

$(x+y)^0$	1	1	$\binom{0}{0}$
$(x+y)^1$	$x + y$	1 1	$\binom{1}{0}\ \binom{1}{1}$
$(x+y)^2$	$x^2 + 2xy + y^2$	1 2 1	$\binom{2}{0}\ \binom{2}{1}\ \binom{2}{2}$
$(x+y)^3$	$x^3 + 3x^2y + 3xy^2 + y^3$	1 3 3 1	$\binom{3}{0}\ \binom{3}{1}\ \binom{3}{2}\ \binom{3}{3}$
$(x+y)^4$	$x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^3$	1 4 6 4 1	$\binom{4}{0}\ \binom{4}{1}\ \binom{4}{2}\ \binom{4}{3}\ \binom{4}{4}$

Pascal's triangle

3. BINOMIAL THEOREM :

The formula by which any positive integral power of a binomial expression can be expanded in the form of a series is known as **BINOMIAL THEOREM.**

If $x, y \in R$ and $n \in N$, then :

$$(x + y)^n = {}^nC_0x^n + {}^nC_1x^{n-1}y + {}^nC_2x^{n-2}y^2 + \ldots + {}^nC_rx^{n-r}y^r + \ldots + {}^nC_ny^n = \sum_{r=0}^{n} {}^nC_rx^{n-r}y^r$$

This theorem can be proved by induction.

Observations :

(a) Symbol $\mathbf{{}^nC_r}$ can also be denoted by $\begin{pmatrix}\mathbf{n}\\ \mathbf{r}\end{pmatrix}$, $\mathbf{C(n, r)}$ or $\mathbf{A_r^n}$.

(b) The number of terms in the expansion is (n+1) i.e. one more than the index.

(c) The sum of the indices of x & y in each term is n.

(d) Power of first variable (x) decreases while of second variable (y) increases.

(e) The binomial coefficients of the terms ($\mathbf{{}^nC_0, {}^nC_1 \ldots}$) equidistant from the beginning and the end are equal. i.e. $\mathbf{{}^nC_r = {}^nC_{n-1}}$

(f) r^{th} term from the beginning in the expansion of $(x + y)^n$ is same as r^{th} term from end in the expansion of $(y + x)^n$.

(g) r^{th} term from the end in $(x + y)^n$ is $\mathbf{(n - r + 2)^{th}}$ **term from the beginning**.

Some important expansions :

(i) $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + ... + {}^nC_nx^n$.

(ii) $(1 - x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 + ... + (-1)^n . {}^nC_nx^n$.

Note : The coefficient of x^r in $(1 + x)^n = {}^nC_r$ & that in $(1-x)^n = (-1)^r . {}^nC_r$

Illustration 1 : Expand : $(y + 2)^6$.

Solution : ${}^6C_0y^6 + {}^6C_1y^5.2 + {}^6C_2y^4.2^2 + {}^6C_3y^3.2^3 + {}^6C_4y^2. 2^4 + {}^6C_5y^1 . 2^5 + {}^6C_6 . 2^6$.

$= y^6 + 12y^5 + 60y^4 + 160y^3 + 240y^2 + 192y + 64$.

Illustration 2 : Write first 4 terms of $\left(1-\frac{2y^2}{5}\right)^7$

Solution : ${}^7C_0, {}^7C_1\left(-\frac{2y^2}{5}\right), {}^7C_2\left(-\frac{2y^2}{5}\right)^2, {}^7C_3\left(-\frac{2y^2}{5}\right)^3$

Illustration 3 : If in the expansion of $(1 + x)^m (1 - x)^n$, the coefficients of x and x^2 are 3 and – 6 respectively then m is - **[JEE 99]**

(A) 6 (B) 9 (C) 12 (D) 24

Solution : $(1 + x)^m (1 - x)^n = \left[1+mx+\frac{(m)(m-1).x^2}{2}+...\right]\left[1-nx+\frac{n(n-1)}{2}x^2+...\right]$

Coefficient of $x = m - n = 3$ ...(i)

Coefficient of $x^2 = -mn + \frac{n(n+1)}{2} + \frac{m(m-1)}{2} = -6$ ...(ii)

Solving (i) and (ii), we get

$m = 12$ and $n = 9$.

Do yourself - 1 :

1. Expand $\left(3x^2-\frac{x}{2}\right)^5$

2. Expand $(y + x)^n$

4. IMPORTANT TERMS IN THE BINOMIAL EXPANSION :

(a) General term: The general term or the $(r+1)^{th}$ term in the expansion of $(x + y)^n$ is given by

$T_{r+1} = {}^nC_r\, x^{n-r}\, y^r$

Illustration 4 : Find : (a) The coefficient of x^7 in the expansion of $\left(ax^2+\frac{1}{bx}\right)^{11}$

(b) The coefficient of x^{-7} in the expansion of $\left(ax-\frac{1}{bx^2}\right)^{11}$

Also, find the relation between a and b, so that these coefficients are equal.

Solution : (a) In the expansion of $\left(ax^2+\frac{1}{bx}\right)^{11}$, the general term is :

$$T_{r+1} = {}^{11}C_r(ax^2)^{11-r}\left(\frac{1}{bx}\right)^r = {}^{11}C_r.\frac{a^{11-r}}{b^r}.x^{22-3r}$$

putting $22 - 3r = 7$

$\therefore \quad 3r = 15 \quad \Rightarrow \quad r = 5$

$\therefore \quad T_6 = {}^{11}C_5\frac{a^6}{b^5}.x^7$

Hence the coefficient of x^7 in $\left(ax^2+\frac{1}{bx}\right)^{11}$ is ${}^{11}C_5a^6b^{-5}$. **Ans.**

Note that binomial coefficient of sixth term is ${}^{11}C_5$.

(b) In the expansion of $\left(ax-\frac{1}{bx^2}\right)^{11}$, general term is :

$$T_{r+1} = {}^{11}C_r(ax)^{11-r}\left(\frac{-1}{bx^2}\right)^r = (-1)^r\,{}^{11}C_r\frac{a^{11-r}}{b^r}.x^{11-3r}$$

putting $11 - 3r = -7$

$\therefore \quad 3r = 18 \quad \Rightarrow \quad r = 6$

$\therefore \quad T_7 = (-1)^6.\ {}^{11}C_6\frac{a^5}{b^6}.x^{-7}$

Hence the coefficient of x^{-7} in $\left(ax-\frac{1}{bx^2}\right)^{11}$ is ${}^{11}C_6a^5b^{-6}$. **Ans.**

Also given :

Coefficient of x^7 in $\left(ax^2+\frac{1}{bx}\right)^{11}$ = coefficient of x^{-7} in $\left(ax-\frac{1}{bx^2}\right)^{11}$

$\Rightarrow \quad {}^{11}C_5a^6b^{-5} = {}^{11}C_6a^5b^{-6}$

$\Rightarrow \quad ab = 1 \qquad (\because\ {}^{11}C_5 = {}^{11}C_6)$

which is the required relation between a and b. **Ans.**

Illustration 5 : Find the number of rational terms in the expansion of $(9^{1/4}+8^{1/6})^{1000}$.

Solution : The general term in the expansion of $(9^{1/4}+8^{1/6})^{1000}$ is

$$T_{r+1} = {}^{1000}C_r\left(9^{\frac{1}{4}}\right)^{1000-r}\left(8^{\frac{1}{6}}\right)^r = {}^{1000}C_r\ 3^{\frac{1000-r}{2}}\ 2^{\frac{r}{2}}$$

The above term will be rational if exponents of 3 and 2 are integers

It means $\frac{1000-r}{2}$ and $\frac{r}{2}$ must be integers

The possible set of values of r is {0, 2, 4, ..., 1000}

Hence, number of rational terms is 501 **Ans.**

(b) Middle term :

The middle term(s) in the expansion of $(x + y)^n$ is (are) :

(i) If n is even, there is only one middle term which is given by $T_{(n+2)/2} = {}^nC_{n/2}.\,x^{n/2}.\,y^{n/2}$

(ii) If n is odd, there are two middle terms which are $T_{(n+1)/2}$ & $T_{[(n+1)/2]+1}$

Important Note :

Middle term has greatest binomial coefficient and if there are 2 middle terms their binomial coefficients will be equal.

(i) ^nC_r will be maximum — When $r=\frac{n}{2}$ if n is even; When $r=\frac{n-1}{2}$ or $\frac{n+1}{2}$ if n is odd

(ii) The term containing greatest binomial coefficient will be middle term in the expansion of $(1+x)^n$

Illustration 6 : Find the middle term in the expansion of $\left(3x-\frac{x^3}{6}\right)^9$

Solution : The number of terms in the expansion of $\left(3x-\frac{x^3}{6}\right)^9$ is 10 (even). So there are two middle terms.

i.e. $\left(\frac{9+1}{2}\right)^{th}$ and $\left(\frac{9+3}{2}\right)^{th}$ are two middle terms. They are given by T_5 and T_6

$$\therefore \quad T_5 = T_{4+1} = {}^9C_4(3x)^5\left(-\frac{x^3}{6}\right)^4 = {}^9C_4 3^5x^5.\frac{x^{12}}{6^4} = \frac{9.8.7.6}{1.2.3.4}.\frac{3^5}{2^4.3^4}x^{17} = \frac{189}{8}x^{17}$$

and $$T_6=T_{5+1}={}^9C_5(3x)^4\left(-\frac{x^3}{6}\right)^5 = -{}^9C_4 3^4.x^4.\frac{x^{15}}{6^5} = \frac{-9.8.7.6}{1.2.3.4}.\frac{3^4}{2^5.3^5}x^{19} = -\frac{21}{16}x^{19}$$ **Ans.**

(c) Term independent of x :

Term independent of x does not contain x ; Hence find the value of r for which the exponent of x is zero.

Illustration 7 : The term independent of x in $\left[\sqrt{\frac{x}{3}}+\sqrt{\left(\frac{3}{2x^2}\right)}\right]^{10}$ is -

(A) 1 (B) $\frac{5}{12}$ (C) $^{10}C_1$ (D) none of these

Solution : General term in the expansion is

$$^{10}C_r\left(\frac{x}{3}\right)^{\frac{r}{2}}\left(\frac{3}{2x^2}\right)^{\frac{10-r}{2}} = {}^{10}C_r x^{\frac{3r}{2}-10}.\frac{3^{5-r}}{2^{\frac{10-r}{2}}}$$ For constant term, $\frac{3r}{2}=10 \Rightarrow r=\frac{20}{3}$

which is not an integer. Therefore, there will be no constant term. **Ans. (D)**

Do yourself - 2 :

1. Find the 7th term of $\left(3x^2-\frac{1}{3}\right)^{10}$

2. Find the term independent of x in the expansion : $\left(2x^2-\frac{3}{x^3}\right)^{25}$

3. Find the middle term in the expansion of :

(a) $\left(\frac{2x}{3}-\frac{3}{2x}\right)^6$; **(b)** $\left(2x^2-\frac{1}{x}\right)^7$

4. Find the term independent of x in the expansion of :

(a) $\left[\sqrt{\frac{x}{3}}+\frac{\sqrt{3}}{2x^2}\right]^{10}$; **(b)** $\left[\frac{1}{2}x^{1/3}+x^{-1/5}\right]^8$

5. Prove that the ratio of the coefficient of x^{10} in $(1-x^2)^{10}$ & the term independent of x in $\left(x-\frac{2}{x}\right)^{10}$ is 1 : 32.

6. Find the coefficient of x^r in the expression :
$(x+3)^{n-1}+(x+3)^{n-2}(x+2)+(x+3)^{n-3}(x+2)^2+\ldots+(x+2)^{n-1}$

7. Element in set of values of r for which, ${}^{18}C_{r-2}+2.\,{}^{18}C_{r-1}+{}^{18}C_r \geq {}^{20}C_{13}$ is :

(A) 9 (B) 5 (C) 7 (D) 10

8. **(a)** If the coefficients of $(2r+4)^{th}$, $(r-2)^{th}$ terms in the expansion of $(1+x)^{18}$ are equal, find r.

(b) If the coefficients of the r^{th}, $(r+1)^{th}$ & $(r+2)^{th}$ terms in the expansion of $(1+x)^{14}$ are in AP, find r.

(c) If the coefficients of 2^{nd}, 3^{rd} & 4^{th} terms in the expansion of $(1+x)^{2n}$ are in AP, show that $2n^2-9n+7=0$.

9. Find the term independent of x in the expansion of $(1+x+2x^3)\left(\frac{3x^2}{2}-\frac{1}{3x}\right)^9$.

10. Let $(1+x^2)^2\,.\,(1+x)^n=\sum_{K=0}^{n+4}a_K\,.\,x^K$. If a_1, a_2 & a_3 are in AP, find n.

11. If n is a positive integer, then $\left(\sqrt{3}+1\right)^{2n}-\left(\sqrt{3}-1\right)^{2n}$ is :

(A) a rational number other than positive integers

(B) an irrational number

(C) an odd positive integer

(D) an even positive integer

(d) Numerically greatest term :

Let numerically greatest term in the expansion of $(a + b)^n$ be T_{r+1}.

$\Rightarrow \quad |T_{r+1}| \geq |T_r|$ and $|T_{r+1}| \geq |T_{r+2}|$ where $T_{r+1} = {}^nC_r a^{n-r} b^r$

Solving above inequalities we get $\dfrac{n+1}{1+\left|\dfrac{a}{b}\right|} - 1 \leq r \leq \dfrac{n+1}{1+\left|\dfrac{a}{b}\right|}$

Case I : When $\dfrac{n+1}{1+\left|\dfrac{a}{b}\right|}$ is an integer equal to m, then T_m and T_{m+1} will be numerically greatest term.

Case II : When $\dfrac{n+1}{1+\left|\dfrac{a}{b}\right|}$ is not an integer and its integral part is m, then T_{m+1} will be the numerically greatest term.

Illustration 8 : Find numerically greatest term in the expansion of $(3 - 5x)^{11}$ when $x = \dfrac{1}{5}$

Solution : Using $\dfrac{n+1}{1+\left|\dfrac{a}{b}\right|} - 1 \leq r \leq \dfrac{n+1}{1+\left|\dfrac{a}{b}\right|}$

$$\frac{11+1}{1+\left|\dfrac{3}{-5x}\right|} - 1 \leq r \leq \frac{11+1}{1+\left|\dfrac{3}{-5x}\right|}$$

solving we get $2 \leq r \leq 3$

$\therefore \quad r = 2, 3$

so, the greatest terms are T_{2+1} and T_{3+1}.

$\therefore \quad$ Greatest term (when $r = 2$)

$T_3 = {}^{11}C_2 . 3^9 (-5x)^2 = 55.3^9 = T_4$

From above we say that the value of both greatest terms are equal. **Ans.**

Illustration 9 : Given T_3 in the expansion of $(1 - 3x)^6$ has maximum numerical value. Find the range of 'x'.

Solution : Using $\dfrac{n+1}{1+\left|\dfrac{a}{b}\right|}-1 \le r \le \dfrac{n+1}{1+\left|\dfrac{a}{b}\right|}$

$$\frac{6+1}{1+\left|\dfrac{1}{-3x}\right|}-1 \le 2 \le \frac{7}{1+\left|\dfrac{1}{-3x}\right|}$$

Let $|x| = t$

$$\frac{21t}{3t+1}-1 \le 2 \le \frac{21t}{3t+1}$$

$$\begin{cases} \dfrac{21t}{3t+1} \le 3 \\ \dfrac{21t}{3t+1} \ge 2 \end{cases} \Rightarrow \begin{cases} \dfrac{4t-1}{3t+1} \le 0 \Rightarrow t \in \left[-\dfrac{1}{3}, \dfrac{1}{4}\right] \\ \dfrac{15t-2}{3t+1} \ge 0 \Rightarrow t \in \left(-\infty, -\dfrac{1}{3}\right] \cup \left[\dfrac{2}{15}, \infty\right) \end{cases}$$

Common solution $t \in \left[\dfrac{2}{15}, \dfrac{1}{4}\right] \Rightarrow x \in \left[-\dfrac{1}{4}, -\dfrac{2}{15}\right] \cup \left[\dfrac{2}{15}, \dfrac{1}{4}\right]$

Do yourself -3 :

1. Find the numerically greatest term in the expansion of $(3 - 2x)^9$, when $x = 1$.

2. In the expansion of $\left(\dfrac{1}{2}+\dfrac{2x}{3}\right)^n$ when $x = -\dfrac{1}{2}$, it is known that 3rd term is the greatest term.

Find the possible integral values of n.

3. Find numerically greatest term in the expansion of :

(a) $(2 + 3x)^9$ when $x = \dfrac{3}{2}$

(b) $(3 - 5x)^{15}$ when $x = \dfrac{1}{5}$

4. For which positive values of x, fourth term in the expansion of $(5 + 3x)^{10}$, is greatest.

5. PROPERTIES OF BINOMIAL COEFFICIENTS :

$$(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + ... + C_nx^n = \sum_{r=0}^{n} {}^nC_rx^r \; ; n \in \mathbb{N} \qquad ....(i)$$

where $C_0, C_1, C_2, ... C_n$ are called combinatorial (binomial) coefficients.

(a) The sum of all the binomial coefficients is 2^n.

Put x = 1, in (i) we get

$$C_0 + C_1 + C_2 + ... + C_n = 2^n \Rightarrow \sum_{r=0}^{n} {}^nC_r = 2^n \qquad ....(ii)$$

(b) Put x = –1 in (i) we get

$$C_0 - C_1 + C_2 - C_3 ... + C_n = 0 \Rightarrow \sum_{r=0}^{n} (-1)^r \, {}^nC_r = 0 \qquad ...(iii)$$

(c) The sum of the binomial coefficients at odd position is equal to the sum of the binomial coefficients at even position and each is equal to 2^{n-1}.

From (ii) & (iii), $C_0 + C_2 + C_4 + ... = C_1 + C_3 + C_5 + ... = 2^{n-1}$

(d) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

(e) $$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

(f) $${}^nC_r = \frac{n}{r} \, {}^{n-1}C_{r-1} = \frac{n}{r}.\frac{n-1}{r-1} \, {}^{n-2}C_{r-2} = ... = \frac{n(n-1)(n-2)...(n-r+1)}{r(r-1)(r-2)...1}$$

(g) $${}^nC_r = \frac{r+1}{n+1}.{}^{n+1}C_{r+1}$$

Illustration 10 : Prove that : ${}^{25}C_{10} + {}^{24}C_{10} + ... + {}^{10}C_{10} = {}^{26}C_{11}$

Solution : LHS $= {}^{10}C_{10} + {}^{11}C_{10} + {}^{12}C_{10} + ... + {}^{25}C_{10}$

$\Rightarrow \; {}^{11}C_{11} + {}^{11}C_{10} + {}^{12}C_{10} + ... + {}^{25}C_{10}$

$\Rightarrow \; {}^{12}C_{11} + {}^{12}C_{10} + ... + {}^{25}C_{10}$

$\Rightarrow \; {}^{13}C_{11} + {}^{13}C_{10} + ... {}^{25}C_{10}$

and so on. $\therefore$ LHS $= {}^{26}C_{11}$

Aliter :

LHS = coefficient of x^{10} in $\{(1 + x)^{10} + (1 + x)^{11} + ... (1+ x)^{25}\}$

$$\Rightarrow \quad \text{coefficient of } x^{10} \text{ in } \left[(1+x)^{10}\frac{\{1+x\}^{16}-1}{1+x-1}\right]$$

$$\Rightarrow \quad \text{coefficient of } x^{10} \text{ in } \frac{\left[(1+x)^{26}-(1+x)^{10}\right]}{x}$$

$$\Rightarrow \quad \text{coefficient of } x^{11} \text{ in } \left[(1+x)^{26}-(1+x)^{10}\right] = {}^{26}C_{11} - 0 = {}^{26}C_{11}$$

Illustration 11 : A student is allowed to select at most n books from a collection of (2n + 1) books. If the total number of ways in which he can select books is 63, find the value of n.

Solution : Given student selects at most n books from a collection of (2n + 1) books. It means that he selects one book or two books or three books or ... or n books. Hence, by the given condition-

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \ldots + {}^{2n+1}C_n = 63 \qquad \ldots(i)$$

But we know that

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \ldots + {}^{2n+1}C_{2n+1} = 2^{2n+1} \qquad \ldots(ii)$$

Since ${}^{2n+1}C_0 = {}^{2n+1}C_{2n+1} = 1$, equation (ii) can also be written as

$$2 + ({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \ldots + {}^{2n+1}C_n) + ({}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + {}^{2n+1}C_{n+3} + \ldots + {}^{2n+1}C_{2n-1} + {}^{2n+1}C_{2n}) = 2^{2n+1}$$

$$\Rightarrow 2 + ({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \ldots + {}^{2n+1}C_n) + ({}^{2n+1}C_n + {}^{2n+1}C_{n-1} + \ldots + {}^{2n+1}C_2 + {}^{2n+1}C_1) = 2^{2n+1}$$

$(\because {}^{2n+1}C_r = {}^{2n+1}C_{2n+1-r})$

$$\Rightarrow 2 + 2({}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \ldots + {}^{2n+1}C_n) = 2^{2n+1} \qquad \text{[from (i)]}$$

$\Rightarrow 2 + 2.63 = 2^{2n+1}$ $\qquad \Rightarrow 1 + 63 = 2^{2n}$

$\Rightarrow 64 = 2^{2n} \Rightarrow 2^6 = 2^{2n}$ $\qquad \therefore 2n = 6$

Hence, n = 3. **Ans.**

Illustration 12 : Prove that :

(i) $C_1 + 2C_2 + 3C_3 + \ldots + nC_n = n \,.\, 2^{n-1}$

(ii) $C_0 + \dfrac{C_1}{2} + \dfrac{C_2}{3} + \ldots + \dfrac{C_n}{n+1} = \dfrac{2^{n+1}-1}{n+1}$

Solution : (i) $\text{L.H.S.} = \sum\limits_{r=1}^{n} r.{}^nC_r = \sum\limits_{r=1}^{n} r.\dfrac{n}{r}.{}^{n-1}C_{r-1}$

$$= n\sum_{r=1}^{n} {}^{n-1}C_{r-1} = n.\left[{}^{n-1}C_0 + {}^{n-1}C_1 + \ldots + {}^{n-1}C_{n-1}\right]$$

$$= n \,.\, 2^{n-1}$$

Aliter : (Using method of differentiation)

$$(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \ldots + {}^nC_nx^n \qquad \ldots(A)$$

Differentiating (A), we get

$$n(1 + x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \ldots + n.C_nx^{n-1}.$$

Put x = 1,

$$C_1 + 2C_2 + 3C_3 + \ldots + n.C_n = n.2^{n-1}$$

(ii) $\text{L.H.S.} = \sum\limits_{r=0}^{n} \dfrac{C_r}{r+1} = \dfrac{1}{n+1}\sum\limits_{r=0}^{n} \dfrac{n+1}{r+1}\,{}^nC_r$

$$= \frac{1}{n+1}\sum_{r=0}^{n} {}^{n+1}C_{r+1} = \frac{1}{n+1}\left[{}^{n+1}C_1 + {}^{n+1}C_2 + \ldots + {}^{n+1}C_{n+1}\right] = \frac{1}{n+1}\left[2^{n+1} - 1\right]$$

Aliter : (Using method of integration)

Integrating (A), we get

$$\frac{(1+x)^{n+1}}{n+1} + C = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \ldots + \frac{C_nx^{n+1}}{n+1} \qquad \text{(where C is a constant)}$$

Put x = 0, we get, $C = -\frac{1}{n+1}$

$$\therefore \quad \frac{(1+x)^{n+1}-1}{n+1} = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + ... + \frac{C_nx^{n+1}}{n+1}$$

Put x = 1, we get $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + ... \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$

Put x = –1, we get $C_0 - \frac{C_1}{2} + \frac{C_2}{3} - ... = \frac{1}{n+1}$

Illustration 13 : If $(1+x)^n = \sum_{r=0}^{n} {}^nC_rx^r$, then prove that $C_1^2 + 2.C_2^2 + 3.C_3^2 + ... + n.C_n^2 = \frac{(2n-1)!}{((n-1)!)^2}$

Solution : $(1 + x)^n = C_0 + C_1x + C_2x^2 + C_2x^3 + ... + C_n\, x^n$ .....(i)

Differentiating both the sides, w.r.t. x, we get

$n(1 + x)^{n-1} = C_1 + 2C_2x + 3C_2x^2 + ... + n.C_nx^{n-1}$ .....(ii)

also, we have

$(x + 1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + ... + C_n$ .....(iii)

Multiplying (ii) & (iii), we get

$(C_1 + 2C_2x + 3C_3x^2 + ... + C_nx^{n-1})(C_0x^n + C_1x^{n-1} + C_2x^{n-2} + ...+ C_n) = n(1 + x)^{2n-1}$

Equating the coefficients of x^{n-1}, we get

$$C_1^2 + 2C_2^2 + 3C_3^2 + ... + n.C_n^2 = n.\,{}^{2n-1}C_{n-1} = \frac{(2n-1)!}{((n-1)!)^2}$$ **Ans.**

Illustration 14 : Prove that : $C_0 - 3C_1 + 5C_2 - ...(-1)^n(2n + 1)C_n = 0$

Solution : $T_r = (-1)^r(2r + 1)\,{}^nC_r = 2(-1)^r r\,.\,{}^nC_r + (-1)^r\,{}^nC_r$

$$\Sigma T_r = 2n\sum_{r=1}^{n}(-1)^r.r.\frac{n}{r}.{}^{n-1}C_{r-1} + \sum_{r=0}^{n}(-1)^r\,{}^nC_r = 2n\sum_{r=1}^{n}(-1)^r.{}^{n-1}C_{r-1} + \sum_{r=0}^{n}(-1)^r.{}^nC_r$$

$$= 2n\left[{}^{n-1}C_0 - {}^{n-1}C_1 + ...\right] + \left[{}^nC_0 - {}^nC_1 + ...\right] = 0$$

Illustration 15 : Prove that $({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - .... + (-1)^{2n}({}^{2n}C_{2n})^2 = (-1)^n.\ {}^{2n}C_n$

Solution : $(1 - x)^{2n} = {}^{2n}C_0 - {}^{2n}C_1x + {}^{2n}C_2x^2 - ....+(-1)^{2n}\,{}^{2n}C_{2n}x^{2n}$ ....(i)

and $(x + 1)^{2n} = {}^{2n}C_0x^{2n} + {}^{2n}C_1x^{2n-1} + {}^{2n}C_2x^{2n-2} +...+{}^{2n}C_{2n}$ ....(ii)

Multiplying (i) and (ii), we get

$(x^2 - 1)^{2n} = ({}^{2n}C_0 - {}^{2n}C_1x + ... + (-1)^{2n}\,{}^{2n}C_{2n}x^{2n}) \times ({}^{2n}C_0x^{2n} + {}^{2n}C_1x^{2n-1} + ... + {}^{2n}C_{2n})$ ....(iii)

Now, coefficient of x^{2n} in R.H.S.

$= ({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - ... + (-1)^{2n}({}^{2n}C_{2n})^2$

$\because$ General term in L.H.S., $T_{r+1} = {}^{2n}C_r(x^2)^{2n-r}(-1)^r$

Putting 2(2n – r) = 2n

$\therefore$ r = n

$\therefore$ $T_{n+1} = {}^{2n}C_nx^{2n}(-1)^n$

Hence coeffiecient of x^{2n} in L.H.S. $= (-1)^n.{}^{2n}C_n$

But (iii) is an identity, therefore coefficient of x^{2n} in R.H.S. = coefficient of x^{2n} in L.H.S.

$\Rightarrow$ $({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - .... + (-1)^n({}^{2n}C_{2n})^2 = (-1)^n.\ {}^{2n}C_n$

Illustration 16 : Prove that : ${}^nC_0.{}^{2n}C_n - {}^nC_1.{}^{2n-2}C_n + {}^nC_2.{}^{2n-4}C_n + ... = 2^n$

Solution : L.H.S. = Coefficient of x^n in $[{}^nC_0(1+x)^{2n} - {}^nC_1(1+x)^{2n-2} ...]$

= Coefficient of x^n in $[(1+x)^2 - 1]^n$

= Coefficient of x^n in $x^n(x+2)^n = 2^n$

Illustration 17 : If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$ then show that the sum of the products of the C_i's taken two at a time represented by : $\sum\limits_{0\le i<j\le n}\sum C_iC_j$ is equal to $2^{2n-1} - \dfrac{2n!}{2.n!n!}$

Solution : Since $(C_0 + C_1 + C_2 + ... + C_{n-1} + C_n)^2$

$= C_0^2 + C_1^2 + C_2^2 + ... + C_{n-1}^2 + C_n^2 + 2(C_0C_1 + C_0C_2 + C_0C_3 + ... + C_0C_n + C_1C_2 + C_1C_3 + ...$
$+ C_1C_n + C_2C_3 + C_2C_4 + ... + C_2C_n + ... + C_{n-1}C_n)$

$(2^n)^2 = {}^{2n}C_n + 2\sum\limits_{0\le i<j\le n}\sum C_iC_j$

Hence $\sum\limits_{0\le i<j\le n}\sum C_iC_j = 2^{2n-1} - \dfrac{2n!}{2.n!n!}$ **Ans.**

Illustration 18 : If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$ then prove that $\sum\limits_{0\le i<j\le n}\sum\left(C_i + C_j\right)^2 = (n-1)\,{}^{2n}C_n + 2^{2n}$

Solution : L.H.S. $\sum\limits_{0\le i<j\le n}\sum\left(C_i + C_j\right)^2$

$= (C_0 + C_1)^2 + (C_0 + C_2)^2 + ... + (C_0 + C_n)^2 + (C_1 + C_2)^2 + (C_1 + C_3)^2 + ...$
$+ (C_1 + C_n)^2 + (C_2 + C_3)^2 + (C_2 + C_4)^2 + ... + (C_2 + C_n)^2 + ... + (C_{n-1} + C_n)^2$

$= n(C_0^2 + C_1^2 + C_2^2 + ... + C_n^2) + 2\sum\limits_{0\le i<j\le n}\sum C_iC_j$

$= n.{}^{2n}C_n + 2.\left\{2^{2n-1} - \dfrac{2n!}{2.n!n!}\right\}$ {from Illustration 17}

$= n.{}^{2n}C_n + 2^{2n} - {}^{2n}C_n = (n-1).{}^{2n}C_n + 2^{2n}$ = R.H.S.

Do yourself - 4 :

1. $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ... + \binom{n}{n} =$

(A) 2^{n-1} (B) ${}^{2n}C_n$ (C) 2^n (D) 2^{n+1}

2. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, $n \in N$. Prove that

(a) $3C_0 - 8C_1 + 13C_2 - 18C_3 + ...$ upto $(n+1)$ terms $= 0$, if $n \ge 2$.

(b) $2C_0 + 2^2\dfrac{C_1}{2} + 2^3\dfrac{C_2}{3} + 2^4\dfrac{C_3}{4} + ... + 2^{n+1}\dfrac{C_n}{n+1} = \dfrac{3^{n+1}-1}{n+1}$

(c) $C_0^2 + \dfrac{C_1^2}{2} + \dfrac{C_2^2}{3} + ... + \dfrac{C_n^2}{n+1} = \dfrac{(2n+1)!}{((n+1)!)^2}$

3. Find the sum of the series $\sum_{r=0}^{n}(-1)^r\,.\,{}^nC_r\left[\frac{1}{2^r}+\frac{3^r}{2^{2r}}+\frac{7^r}{2^{3r}}+\frac{15^r}{2^{4r}}+\ldots\text{up to m terms}\right]$

4. Prove the following identities using the theory of permutation where $C_0, C_1, C_2, \ldots, C_n$ are the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in N$:

(a) $C_0^2 + C_1^2 + C_2^2 + \ldots + C_n^2 = \frac{(2n)!}{n!\,n!}$

(b) $C_0C_1 + C_1C_2 + C_2C_3 + \ldots + C_{n-1}C_n = \frac{(2n)!}{(n+1)!\,(n-1)!}$

(c) $C_oC_r + C_1C_{r+1} + C_2C_{r+2} + \ldots + C_{n-r}C_n = \frac{2n!}{(n-r)!\,(n+r)!}$

(d) $\sum_{r=0}^{n-2}({}^nC_r\,.\,{}^nC_{r+2}) = \frac{(2n)!}{(n-2)!(n+2)!}$

(e) ${}^{100}C_{10} + 5.\,{}^{100}C_{11} + 10\,.\,{}^{100}C_{12} + 10\,.\,{}^{100}C_{13} + 5.\,{}^{100}C_{14} + {}^{100}C_{15} = {}^{105}C_{90}$

5. If $C_0, C_1, C_2, \ldots, C_n$ are the combinatorial coefficients in the expansion of $(1+x)^n$, $n \in N$, then prove the following :

(a) $C_1 + 2C_2 + 3C_3 + \ldots + n\,.\,C_n = n\,.\,2^{n-1}$

(b) $C_0 + 2C_1 + 3C_2 + \ldots + (n+1)C_n = (n+2)2^{n-1}$

(c) $C_0 + 3C_1 + 5C_2 + \ldots + (2n+1)C_n = (n+1)\,2^n$

(d) $(C_0 + C_1)(C_1 + C_2)(C_2 + C_3)\ldots(C_{n-1} + C_n) = \frac{C_0.C_1.C_2\ldots C_{n-1}(n+1)^n}{n!}$

(e) $1\,.\,C_o^2 + 3\,.\,C_1^2 + 5\,.\,C_2^2 + \ldots + (2n+1)\,C_n^2 = \frac{(n+1)\,(2n)!}{n!\,n!}$

6. Prove that

(a) $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \ldots + \frac{n.C_n}{C_{n-1}} = \frac{n(n+1)}{2}$

(b) $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$

(c) $2\,.\,C_o + \frac{2^2.C_1}{2} + \frac{2^3.C_2}{3} + \frac{2^4.C_3}{4} + \ldots \frac{2^{n+1}.C_n}{n+1} = \frac{3^{n+1}-1}{n+1}$

(d) $C_o - \frac{C_1}{2} + \frac{C_2}{3} - \ldots + (-1)^n\frac{C_n}{n+1} = \frac{1}{n+1}$

7. Prove that $\sum_{K=0}^{n} {}^{n}C_K \sin Kx.\cos(n-K)x = 2^{n-1}\sin nx$.

8. If $\binom{n}{r}$ denotes ${}^{n}C_r$, then

(a) Evaluate : $2^{15}\binom{30}{0}\binom{30}{15} - 2^{14}\binom{30}{1}\binom{29}{14} + 2^{13}\binom{30}{2}\binom{28}{13} \ldots - \binom{30}{15}\binom{15}{0}$

(b) Prove that : $\sum_{r=1}^{n}\binom{n-1}{n-r}\binom{n}{r} = \binom{2n-1}{n-1}$

(c) Prove that : $\binom{n}{r}\binom{r}{k} = \binom{n}{k}\binom{n-k}{r-k}$

9. Let $S_1 = \sum_{j=1}^{10} j(j-1)\,{}^{10}C_j$, $S_2 = \sum_{j=1}^{10} j\,{}^{10}C_j$ and $S_3 = \sum_{j=1}^{10} j^2\,{}^{10}C_j$.

Statement–1 : $S_3 = 55 \times 2^9$.

Statement–2 : $S_1 = 90 \times 2^8$ and $S_2 = 10 \times 2^8$.

(A) Statement–1 is true, Statement–2 is true ; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is true, Statement–2 is true ; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is true, Statement–2 is false.

(D) Statement–1 is false, Statement–2 is true.

10. For r = 0, 1, ... , 10, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r(B_{10}B_r - C_{10}A_r)$ is equal to -

(A) $B_{10} - C_{10}$ (B) $A_{10}\left(B_{10}^2 - C_{10}A_{10}\right)$

(C) 0 (D) $C_{10} - B_{10}$

11. Let $N = {}^{2000}C_1 + 2 \cdot {}^{2000}C_2 + 3 \cdot {}^{2000}C_3 + \ldots + 2000 \cdot {}^{2000}C_{2000}$. Prove that N is divisible by 2^{2003}.

6. MULTINOMIAL THEOREM :

Using binomial theorem, we have $(x+a)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r} a^r,\ n \in N$

$$= \sum_{r=0}^{n} \frac{n!}{(n-r)!r!} x^{n-r} a^r = \sum_{r+s=n} \frac{n!}{r!s!} x^s a^r \text{, where } s+r=n$$

This result can be generalized in the following form.

$$(x_1 + x_2 + \ldots + x_k)^n = \sum_{r_1+r_2+\ldots+r_k=n} \frac{n!}{r_1!r_2!\ldots r_k!} x_1^{r_1} x_2^{r_2} \ldots x_k^{r_k}$$

The general term in the above expansion $\frac{n!}{r_1!r_2!r_3!\ldots r_k!} \cdot x_1^{r_1} x_2^{r_2} x_3^{r_3} \ldots x_k^{r_k}$

The number of terms in the above expansion is equal to the number of non-negative integral solution of the equation $r_1 + r_2 + \ldots + r_k = n$ because each solution of this equation gives a term in the above expansion. The number of such solutions is ${}^{n+k-1}C_{k-1}$

Particular cases :

(i) $(x+y+z)^n = \sum_{r+s+t=n} \frac{n!}{r!s!t!} x^r y^s z^t$

The above expansion has ${}^{n+3-1}C_{3-1} = {}^{n+2}C_2$ terms

(ii) $(x+y+z+u)^n = \sum_{p+q+r+s=n} \frac{n!}{p!q!r!s!} x^p y^q z^r u^s$

There are ${}^{n+4-1}C_{4-1} = {}^{n+3}C_3$ terms in the above expansion.

Illustration 19 : Find the coefficient of $x^2 y^3 z^4 w$ in the expansion of $(x-y-z+w)^{10}$

Solution : $(x-y-z+w)^{10} = \sum_{p+q+r+s=10} \frac{n!}{p!q!r!s!} (x)^p (-y)^q (-z)^r (w)^s$

We want to get $x^2y^3z^4w$ this implies that $p = 2, q = 3, r = 4, s = 1$

$\therefore$ Coefficient of $x^2y^3z^4w$ is $\frac{10!}{2!\,3!\,4!\,1!}(-1)^3(-1)^4 = -12600$ **Ans.**

Illustration 20 : Find the total number of terms in the expansion of $(1+x+y)^{10}$ and coefficient of x^2y^3.

Solution : Total number of terms $= {}^{10+3-1}C_{3-1} = {}^{12}C_2 = 66$

Coefficient of $x^2y^3 = \frac{10!}{2! \times 3! \times 5!} = 2520$ **Ans.**

Illustration 21 : Find the coefficient of x^5 in the expansion of $(2 - x + 3x^2)^6$.

Solution : The general term in the expansion of $(2-x+3x^2)^6 = \dfrac{6!}{r!s!t!}2^r(-x)^s(3x^2)^t$,

where $r + s + t = 6$.

$= \dfrac{6!}{r!s!t!}2^r \times (-1)^s \times (3)^t \times x^{s+2t}$

For the coefficient of x^5, we must have $s + 2t = 5$.

But, $r + s + t = 6$,

$\therefore \quad s = 5 - 2t$ and $r = 1 + t$, where $0 \le r, s, t \le 6$.

Now $t = 0 \Rightarrow r = 1,\ s = 5.$

$t = 1 \Rightarrow r = 2,\ s = 3.$

$t = 2 \Rightarrow r = 3,\ s = 1.$

Thus, there are three terms containing x^5 and coefficient of x^5

$$= \frac{6!}{1!\,5!\,0!}\times 2^1 \times (-1)^5 \times 3^0 + \frac{6!}{2!\,3!\,1!}\times 2^2 \times (-1)^3 \times 3^1 + \frac{6!}{3!\,1!\,2!}\times 2^3 \times (-1)^1 \times 3^2$$

$= -12 - 720 - 4320 = -5052.$ **Ans.**

Illustration 22 : If $(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then prove that (a) $a_r = a_{2n-r}$ (b) $\sum_{r=0}^{n-1} a_r = \frac{1}{2}(3^n - a_n)$

Solution : (a) We have

$$\left(1+x+x^2\right)^n = \sum_{r=0}^{2n} a_r x^r \qquad ....(A)$$

Replace x by $\dfrac{1}{x}$

$$\therefore \quad \left(1+\frac{1}{x}+\frac{1}{x^2}\right)^n = \sum_{r=0}^{2n} a_r\left(\frac{1}{x}\right)^r$$

$$\Rightarrow \quad \left(x^2+x+1\right)^n = \sum_{r=0}^{2n} a_r x^{2n-r}$$

$$\Rightarrow \quad \sum_{r=0}^{2n} a_r x^r = \sum_{r=0}^{2n} a_r x^{2n-r} \qquad \{\text{Using (A)}\}$$

Equating the coefficient of x^{2n-r} on both sides, we get

$a_{2n-r} = a_r$ for $0 \le r \le 2n$.

Hence $a_r = a_{2n-r}$.

(b) Putting x=1 in given series, then

$a_0 + a_1 + a_2 + ...+ a_{2n} = (1+1+1)^n$

$a_0 + a_1 + a_2 + ...+ a_{2n} = 3^n \qquad ....(1)$

But $a_r = a_{2n-r}$ for $0 \le r \le 2n$

$\therefore$ series (1) reduces to

$2(a_0 + a_1 + a_2 + ...+ a_{n-1}) + a_n = 3^n.$

$\therefore \quad a_0 + a_1 + a_2 + ...+ a_{n-1} = \dfrac{1}{2}(3^n - a_n)$

Do yourself - 5 :

1. Find the coefficient of x^2y^5 in the expansion of $(3+2x-y)^{10}$.

2. Find the coefficient of

(a) x^4 in the expansion of $(1+x+x^2+x^3)^{11}$

(b) x^4 in the expansion of $(2-x+3x^2)^6$

3. Given that $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$, find the values of :

(i) $a_0 + a_1 + a_2 + \ldots + a_{2n}$;

(ii) $a_0 - a_1 + a_2 - a_3 \ldots + a_{2n}$;

(iii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots + a_{2n}^2$

4. Find the coefficient of

(a) $x^2y^3z^4$ in the expansion of $(ax-by+cz)^9$.

(b) $a^2b^3c^4d$ in the expansion of $(a-b-c+d)^{10}$.

7. APPLICATION OF BINOMIAL THEOREM :

Illustration 23 : If $\left(6\sqrt{6}+14\right)^{2n+1} = [N] + F$ and $F = N - [N]$; where [.] denotes greatest integer function, then NF is equal to

(A) 20^{2n+1} (B) an even integer (C) odd integer (D) 40^{2n+1}

Solution : Since $\left(6\sqrt{6}+14\right)^{2n+1} = [N] + F$

Let us assume that $f = \left(6\sqrt{6}-14\right)^{2n+1}$; where $0 \le f < 1$.

Now, $[N] + F - f = \left(6\sqrt{6}+14\right)^{2n+1} - \left(6\sqrt{6}-14\right)^{2n+1}$

$$= 2\left[{}^{2n+1}C_1\left(6\sqrt{6}\right)^{2n}(14) + {}^{2n+1}C_3\left(6\sqrt{6}\right)^{2n-2}(14)^3 + \ldots\right]$$

$\Rightarrow$ $[N] + F - f =$ even integer.

Now $0 < F < 1$ and $0 < f < 1$

so $-1 < F - f < 1$ and $F - f$ is an integer so it can only be zero

Thus $NF = \left(6\sqrt{6}+14\right)^{2n+1}\left(6\sqrt{6}-14\right)^{2n+1} = 20^{2n+1}$. **Ans. (A,B)**

Illustration 24 : Find the last three digits in 11^{50}.

Solution : Expansion of $(10+1)^{50} = {}^{50}C_0 10^{50} + {}^{50}C_1 10^{49} + \ldots + {}^{50}C_{48} 10^2 + {}^{50}C_{49} 10 + {}^{50}C_{50}$

$$= \underbrace{{}^{50}C_0 10^{50} + {}^{50}C_1 10^{49} + \ldots + {}^{50}C_{47} 10^3}_{1000K} + 49 \times 25 \times 100 + 500 + 1$$

$\Rightarrow$ $1000\,K + 123001$

$\Rightarrow$ Last 3 digits are 001.

Illustration 25 : Prove that $2222^{5555} + 5555^{2222}$ is divisible by 7.

Solution : When 2222 is divided by 7 it leaves a remainder 3.

So adding & subtracting 3^{5555}, we get :

$$E = \underbrace{2222^{5555} - 3^{5555}}_{E_1} + \underbrace{3^{5555} + 5555^{2222}}_{E_2}$$

For E_1 : Now since $2222 - 3 = 2219$ is divisible by 7, therefore E_1 is divisible by 7

($\because$ $x^n - a^n$ is divisible by $x - a$)

For E_2 : 5555 when devided by 7 leaves remainder 4.

So adding and subtracting 4^{2222}, we get :

$$E_2 = 3^{5555} + 4^{2222} + 5555^{2222} - 4^{2222}$$

$$= (243)^{1111} + (16)^{1111} + (5555)^{2222} - 4^{2222}$$

Again $(243)^{1111} + 16^{1111}$ and $(5555)^{2222} - 4^{2222}$ are divisible by 7

($\because$ $x^n + a^n$ is divisible by $x + a$ when n is odd)

Hence $2222^{5555} + 5555^{2222}$ is divisible by 7.

Do yourself - 6 :

1. Prove that $5^{25} - 3^{25}$ is divisible by 2.

2. Find the remainder when the number 9^{100} is divided by 8.

3. Find last three digits in 19^{100}.

4. Let $R = (8+3\sqrt{7})^{20}$ and [.] denotes greatest integer function, then prove that :

(a) [R] is odd (b) $R - [R] = 1 - \dfrac{1}{(8+3\sqrt{7})^{20}}$

5. Find the digit at unit's place in the number $17^{1995} + 11^{1995} - 7^{1995}$.

6. (a) Show that the integral part in each of the following is odd. $n \in N$

(A) $\left(5+2\sqrt{6}\right)^n$ (B) $\left(8+3\sqrt{7}\right)^n$

(b) Show that the integral part in each of the following is even. $n \in N$

(A) $\left(3\sqrt{3}+5\right)^{2n+1}$ (B) $\left(5\sqrt{5}+11\right)^{2n+1}$

7. (i) Find the remainder when 7^{98} is divided by 5

(ii) Using binomial theorem prove that $6^n - 5n$ always leaves the remainder 1 when divided by 25.

(iii) Find the last digit, last two digits and last three digits of the number $(27)^{27}$.

8. If {x} denotes the fractional part of 'x', then $82\left\{\dfrac{3^{1001}}{82}\right\} =$

8. BINOMIAL THEOREM FOR NEGATIVE OR FRACTIONAL INDICES :

If $n \in \mathbb{R}$, then $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots \infty$ provided $|x| < 1$.

Note :

(i) When the index n is a positive integer the number of terms in the expansion of $(1+x)^n$ is finite i.e. $(n+1)$ & the coefficient of successive terms are : ${}^nC_0, {}^nC_1, {}^nC_2, \ldots {}^nC_n$

(ii) When the index is other than a positive integer such as negative integer or fraction or real number, the number of terms in the expansion of $(1+x)^n$ is infinite.

(iii) Following expansion should be remembered ($|x| < 1$).

(a) $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \ldots \infty$

(b) $(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \ldots \infty$

(c) $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots \infty$

(d) $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots \infty$

(e) $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \ldots + \frac{(-1)^r(r+1)(r+2)}{2!}x^r + \ldots$

(f) $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \ldots + \frac{(r+1)(r+2)}{2!}x^r + \ldots$

(iv) The expansions in ascending powers of x are only valid if x is 'small'. If x is large i.e. $|x| > 1$ then we may find it convenient to expand in powers of $1/x$, which then will be small.

9. APPROXIMATIONS :

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1.2}x^2 + \frac{n(n-1)(n-2)}{1.2.3}x^3 \ldots$$

If $|x| < 1$, the terms of the above expansion go on decreasing and if x be very small, a stage may be reached when we may neglect the terms containing higher powers of x in the expansion. Thus, if x be so small that its square and higher powers may be neglected then $(1+x)^n = 1 + nx$, approximately.

This is an approximate value of $(1+x)^n$

Illustration 26 : If x is so small such that its square and higher powers may be neglected then find the approximate value of $\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}}$

Solution :

$$\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}} = \frac{1 - \frac{3}{2}x + 1 - \frac{5x}{3}}{2\left(1+\frac{x}{4}\right)^{1/2}} = \frac{1}{2}\left(2 - \frac{19}{6}x\right)\left(1+\frac{x}{4}\right)^{-1/2} = \frac{1}{2}\left(2 - \frac{19}{6}x\right)\left(1 - \frac{x}{8}\right)$$

$$= \frac{1}{2}\left(2 - \frac{x}{4} - \frac{19}{6}x\right) = 1 - \frac{x}{8} - \frac{19}{12}x = 1 - \frac{41}{24}x$$ **Ans.**

Illustration 27 : The value of cube root of 1001 upto five decimal places is –

(A) 10.03333 (B) 10.00333 (C) 10.00033 (D) none of these

Solution : $(1001)^{1/3} = (1000+1)^{1/3} = 10\left(1+\frac{1}{1000}\right)^{1/3} = 10\left\{1+\frac{1}{3}\cdot\frac{1}{1000}+\frac{1/3(1/3-1)}{2!}\frac{1}{1000^2}+\ldots\right\}$

$= 10\{1 + 0.0003333 - 0.00000011 + \ldots\} = 10.00333$ **Ans. (B)**

Illustration 28 : The sum of $1+\frac{1}{4}+\frac{1.3}{4.8}+\frac{1.3.5}{4.8.12}+\ldots\infty$ is -

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{3}$ (D) $2^{3/2}$

Solution : Comparing with $1+nx+\frac{n(n-1)}{2!}x^2+\ldots$

$nx = 1/4$ ....(i)

and $\frac{n(n-1)x^2}{2!}=\frac{1.3}{4.8}$

or $\frac{nx(nx-x)}{2!}=\frac{3}{32} \Rightarrow \frac{1}{4}\left(\frac{1}{4}-x\right)=\frac{3}{16}$ (by (i))

$\Rightarrow \left(\frac{1}{4}-x\right)=\frac{3}{4} \Rightarrow x=\frac{1}{4}-\frac{3}{4}=-\frac{1}{2}$ ....(ii)

putting the value of x in (i)

$n(-1/2) = 1/4 \Rightarrow n = -1/2$

$\therefore$ sum of series $= (1+x)^n = (1-1/2)^{-1/2} = (1/2)^{-1/2} = \sqrt{2}$ **Ans. (A)**

10. EXPONENTIAL SERIES :

(a) e is an irrational number lying between 2.7 & 2.8. Its value correct upto 10 places of decimal is 2.7182818284.

(b) Logarithms to the base 'e' are known as the Napierian system, so named after Napier, their inventor. They are also called **Natural Logarithm.**

(c) $e^x = 1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots\infty$; where x may be any real or complex number & $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$

(d) $a^x = 1+\frac{x}{1!}\ell na+\frac{x^2}{2!}\ell n^2 a+\frac{x^3}{3!}\ell n^3 a+\ldots\infty$, where $a>0$

(e) $e = 1+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\ldots\infty$

11. LOGARITHMIC SERIES :

(a) $\ell n\,(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\infty$, where $-1 < x \le 1$

(b) $\ell n\,(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + \ldots\infty$, where $-1 \le x < 1$

Remember : (i) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots\infty = \ell n\,2$ (ii) $e^{\ell nx} = x$; for all $x > 0$

(iii) $\ell n2 = 0.693$ (iv) $\ell n10 = 2.303$

Do yourself - 7 :

1. Find the co-efficient of x^6 in the expansion of $(1 - 2x)^{-5/2}$.

2. (i) Find the coefficient of x^{12} in $\frac{4+2x-x^2}{(1+x)^3}$

(ii) Find the coefficient of x^{100} in $\frac{3-5x}{(1-x)^2}$

3. If $|x| < 1$, then the co-efficient of x^n in the expansion of $(1 + x + x^2 + x^3 + \ldots)^2$ is

(A) n (B) n – 1 (C) n + 2 (D) n + 1

4. The coefficient of x^4 in $\left(\frac{1+x}{1-x}\right)^2$, $|x| < 1$, is

(A) 4 (B) –4 (C) $10 + {}^4C_2$ (D) 16